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Data-Driven Koopman-based Modeling and Control of Quadcopter in a Turbulent Wind Field

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Abstract— **We present a data-driven approach to model wind effects on quadcopter trajectory tracking and make use of Model Predictive Control (MPC) to improve the tracking performance in a turbulent wind field. To generate datasets for learning, we employ a quadcopter dynamic model enhanced with additional terms that incorporate wind effects on body and rotor dynamics. We propose a tracking error-based learning model based on the Koopman operator theory. We collect the data with an existing controller in the loop and design MPC controllers to augment the existing controller. The Koopman-based model captures the nonlinear dynamics in quadcopter trajectory tracking in a linear form. We design two different MPC controllers and evaluate their performance using Monte Carlo simulations. The controllers differ in how the wind is modeled and predicted. We demonstrate that these controllers improve tracking performance and reduce tracking errors better.**

I. INTRODUCTION

The global endeavor to increase public confidence and comfort in the integration of passenger and cargo aerial transportation in urban, suburban, rural, and regional areas has posed technical challenges to the guidance, navigation, and control community. Low-altitude missions create different operational challenges due to high-density structural environments and atmospheric phenomena. The performance of the operations largely depends on the knowledge of accurate dynamics of aerial vehicles, especially for small uncrewed aircraft systems such as quadcopters. In the presence of uncertainty such as wind, accurate modeling of aeromechanics and aerodynamics is crucial for controller design and performance. Disturbance rejection controllers have been used to reduce wind effects, either by treating it as an unknown disturbance or by utilizing prediction and estimation methods to find disturbance. However, the turbulence effect is not captured by simplified quadcopter and rotor dynamics [1].

Data-driven modeling is a popular and efficient approach adopted for several platforms such as ground robots [2, 3, 4], cars [5, 6] as well as aerial vehicles [7, 8, 9, 10, 11, 12]. This line of work either adopts fully data-driven modeling or partial dynamics correction modeling through neural network (NN) [4, 5, 3], Gaussian process (GP) [6, 8, 9] and Koopman operator [12, 11]. Recently, Koopman-based modeling and identification have received enormous attention from researchers. In 1931, Bernard O. Koopman [13] demonstrated that it is possible to represent a nonlinear dynamical system in terms of an infinite-dimensional linear operator acting on a Hilbert space. This operator, named the Koopman operator, is linear, and its spectral decomposition completely characterizes the behavior of a nonlinear system. The appeal and utility of expressing nonlinear dynamics in a linear framework are vast, considering easy-to-use linear control designs and state estimation techniques. Although Koopman Operator is infinite-dimensional, it is possible to find a finite-dimensional approximation of the operator.

Studies have been carried out to incorporate aerodynamic effects such as body drag [7, 8], and rotor drag [14, 1, 15] into aerial vehicle dynamics. Data-driven methods [8, 10, 14] have considered the effects of wind as a disturbance force. References [8, 10] employ Gaussian process regression to model wind effects on the rotor. Wind is modeled as a radial basis function while generating the data in [8]. Thrust and torque residual effects are considered in [14] with hybrid modeling. Specifically, the authors propose a high-fidelity model and implement neural networks along with the first principle method.

In this paper, we focus on capturing wind effects on quadcopter operations and improving tracking performance. We propose to learn a dynamic model of the trajectory deviation due to wind effects. Towards this end, we consider a quadcopter platform with a tracking controller onboard. The quadcopter dynamics include a drag model and a high-fidelity rotor dynamics model derived from the Blade Element Momentum Theory [1]. The onboard controller is used to track reference trajectories. With the onboard controller, we generate trajectory tracking data in the presence of wind to capture its effect on the tracking performance.

We propose two different data collection methods. The first method collects trajectory tracking errors, random perturbations added to the onboard controller, and wind velocity. The second method additionally collects a residual force, which is the difference between the commanded thrust and an estimate of the executed thrust. We assume that the rotor speed of the quadcopter is available through sensor measurements and that the commanded rotor speed is also available. Using these two pieces of information, we compute the residual force and integrate it into the modeling of the error dynamics.

Using Koopman operator theory, we create linear dynamic models from wind velocity, trajectory tracking error, and control inputs. The finite-dimensional approximation of the

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Koopman operator is achieved by minimizing the l_1 regularization with extended dynamic mode decomposition [15]. To reduce the tracking error, we leverage the learned Koopman model and augment the existing controller with MPC.

We evaluate two different MPC formulations in simulations. We find that both formulations improve the tracking performance and can reduce the error by as much as 31%. We also find that incorporating the residual force results in a significant improvement in tracking error reduction.

The rest of this paper is organized as follows. In Section II, we discuss improved quadcopter dynamics including high-fidelity rotor dynamics. Section III describes the proposed solution, including model identification and MPC. In Section IV we present simulation results. Conclusions and future work are discussed in Section V.

II. DYNAMIC MODELS OF A QUADCOPTER

A. Quadcopter Modeling in Wind

We consider a quadcopter aerial vehicle as a single rigid body with four identical rotors. A 13-dimensional state-space model of this quadcopter includes $\boldsymbol{p} = [p_x, p_y, p_z] \in \mathbb{R}^3$ the inertial position, $\boldsymbol{q} = [q_w, q_x, q_y, q_z]^T \in \mathbb{R}^4$ the unit quaternion representing its orientation in the inertial frame, $\boldsymbol{v} = [v_x, v_y, v_z] \in \mathbb{R}^3$ the inertial velocity, and $\boldsymbol{\omega} \in \mathbb{R}^3$ the angular rate represented in the body frame. The nominal quadcopter dynamics can be found in [16, 1]. To include wind effects, we update the quadrotor model to

$$\dot{\boldsymbol{p}} = \boldsymbol{v_r} + \boldsymbol{v_w} \tag{1}$$

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} \tag{2}$$

$$\dot{\boldsymbol{v_r}} = \boldsymbol{g} + \frac{1}{m}\boldsymbol{q} \odot \boldsymbol{f_t} - \frac{1}{m}\boldsymbol{f_D} \tag{3}$$

$$\dot{\boldsymbol{\omega}} = \boldsymbol{J}^{-1} \cdot \boldsymbol{\tau_t} - \boldsymbol{J}^{-1}[\boldsymbol{\omega} \times (J \cdot \boldsymbol{\omega})] \tag{4}$$

where $\odot$ and $\otimes$ are the quaternion rotation and multiplication, $\boldsymbol{g} = [\,0;\ 0;\ g]^T$ represents the gravitational acceleration, m is the mass, $\boldsymbol{J}$ is the moment of inertia of the quadcopter, $\boldsymbol{f_D} \in \mathbb{R}^3$ is the drag force on the quadcopter in the inertial frame, and $\boldsymbol{f_t}, \boldsymbol{\tau_t} \in \mathbb{R}^3$ are the thrust and torque, respectively, which include the wind effect on the rotors. Let $\boldsymbol{\Omega} = [\Omega_1,\ \Omega_2,\ \Omega_3,\ \Omega_4]^T$ be the propeller speed expressed in Revolution Per Minute (RPM). The simplest rotor model is a quadratic fitting model that assumes the thrust and the torque produced by a single rotor are proportional to the square of the rotational speed of the propeller. To estimate the drag, we use a quadratic drag model of the following form $\boldsymbol{f_D} = \boldsymbol{R}\boldsymbol{D}\boldsymbol{v_r^b}||\boldsymbol{v_r^b}||$, where $\boldsymbol{R} \in SO(3)$ is the orientation matrix represented by $\boldsymbol{q}$, and $\boldsymbol{v_r^b} = \boldsymbol{R}^T\boldsymbol{v_r}$ is the relative air velocity in the body frame and $\boldsymbol{D}$ is a diagonal matrix containing the drag coefficients.

We model the wind velocity in the inertial frame, $\boldsymbol{v_w} \in \mathbb{R}^3$, as the summation of a mean component ($\bar{\boldsymbol{v}}_{\boldsymbol{w}}$) and a stochastic component ($\tilde{\boldsymbol{v}}_{\boldsymbol{w}}$), that is, $\boldsymbol{v_w} = \bar{\boldsymbol{v}}_{\boldsymbol{w}} + \tilde{\boldsymbol{v}}_{\boldsymbol{w}}$. Formulation of the turbulence flow as a combination of a mean velocity component and an intensity or variance component is present in the engineering application [17]. Stochastic formulations of $\tilde{\boldsymbol{v}}_{\boldsymbol{w}}$ like [18] and its variants are majorly dependent on canonical spectral energy function for incorporating disturbances or gusts in the wind field. To simplify the formulation, we model $\tilde{\boldsymbol{v}}_{\boldsymbol{w}}$ as a zero-mean Gaussian noise.

To incorporate rotor effects, we integrate a hybrid Blade Element Momentum theory (BEMT) [1] that accounts for the effects of relative air speed on the rotor thrust. Momentum theory considers the rotor as an actuator disk within a flow tube and the thrust is derived assuming a uniform inflow ratio, λ_0. On the other hand, Blade Element Theory (BET) accounts for the infinitesimal blade element to sum up its contribution to thrust and torque. Here the inflow ratio distribution across the rotor is non-uniform. The radial, tangential, and perpendicular velocities of an individual blade element can be non-dimensionalized by the blade tip velocity. The hybrid BEMT seeks to optimize a candidate set of inflow ratio and thrust coefficients (λ_0, b) that satisfies both momentum theory and BET. The algorithm to find the adjusted thrust and associated rotor angular velocity is from [1]. Due to the rotor interaction with wind, the executed thrust as well as the rotor angular velocity $\boldsymbol{\Omega}$ can be different from the commanded thrust and RPM. The commanded RPM is calculated from the thrust and the torque using the quadratic rotor model and the adjusted RPM is estimated from the BEMT algorithm.

III. ERROR MODEL LEARNING AND IMPROVED CONTROL WITH KOOPMAN OPERATOR

A. Overview of the Proposed Solution

A few recent studies [12, 11, 19, 20, 21] have explored identifying quadcopter dynamics using the Koopman operator theory. Koopman-based controllers have been proven to be better at tracking than controllers based on a linearized model and nonlinear model predictive controllers in some cases [11]. Although some of the studies propose robust identification processes, the quadcopter dynamics considered are basic. Most of the models were fit without considering environmental uncertainty. For example, the datasets generated from the experimental platform consider nominal decoupled dynamics for planer trajectories [12]. The simulated datasets generated do not account for the complex wind-rotor interaction effects [20]. In the presence of wind, a nominal controller onboard the quadcopter may not perform satisfactorily due to the additional effects from the wind, such as body drag, rotor drag, and effects on thrust inputs. Hence, without the additional knowledge of the true dynamics, the controller performance degrades.

The objective of this paper is to reduce quadcopter tracking errors while operating in a turbulent wind field. To capture the wind effect on the quadcopter dynamics holistically, we propose a data-driven approach to learning a dynamic model that takes perturbations of existing control signals as the control input, wind velocities as the disturbance input, and the trajectory tracking error as the output. We employ the Koopman operator theory and learn the model in

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the lifted state space. The learned model is further used in MPC to reduce the trajectory tracking error.

Fig. 1 shows the data collection process. We consider a quadcopter with an existing high-level controller and a low-level controller. In this paper, the high-level control is an LQR tracking controller [16] that outputs desired angular velocity and thrust $\boldsymbol{u_h} = [\omega_x, \omega_y, \omega_z, f_c]^T \in \mathbb{R}^4$ for the low-level control to track. The LQR controller may be designed without consideration of wind. Therefore, its performance against wind may not be satisfactory. Nevertheless, we assume that the LQR control ensures relatively stable flight as the quadcopter flies in a windy environment.

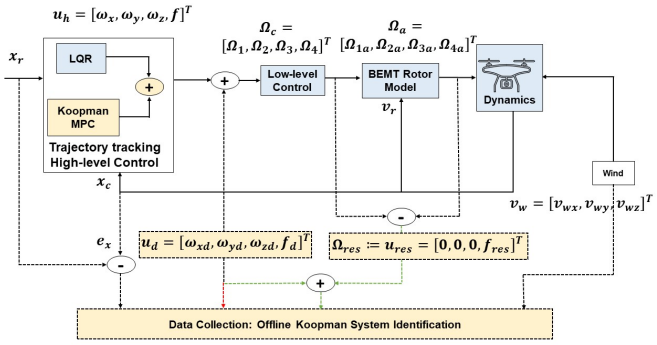


Fig. 1. A diagram of the system structure, the data collection flow, and controller augmentation. The blocks in blue accent show the current system architecture, and the blocks in yellow accent represent the system structure for Koopman-based identification and control. The dashed line indicates the data collection process, and the solid line block in the high-level control block represents the augmentation with the current system. The state, random perturbations on the input, residual input, and wind data are collected at 10 Hz. For model 1 and model 2, the state data are the same. The data flow in red represents the input for model 1 and the flow in green represents the data collection flow of input for model 2.

We employ the updated quadcopter dynamics and high-fidelity rotor dynamics in Section II-A to generate datasets for model identification. Specifically, the state data used in the modeling are the 3-dimensional position, velocity, and orientation (roll, pitch, yaw) data. We add random perturbations $\boldsymbol{u_d} = [\omega_{xd}, \omega_{yd}, \omega_{zd}, f_d]^T$ to the high-level control $\boldsymbol{u_h}$. We then collect the state data to identify a dynamic model from the perturbations $\boldsymbol{u_d}$ and the wind $\boldsymbol{v_w}$. The trajectory tracking error $\boldsymbol{e_x}$ is calculated from the state $\boldsymbol{x_c}$ and reference $\boldsymbol{x_r}$ data. The tracking error $\boldsymbol{e_x}$ is then lifted with a polynomial basis function. Denote by $\boldsymbol{\psi_k^e}$ and $\boldsymbol{v_{w_k}}$ the *lifted* trajectory tracking error and the wind velocity at time step k, respectively. We assume that the identified model takes the form

$$\boldsymbol{\psi_{k+1}^e} = \boldsymbol{A\psi_k^e} + \boldsymbol{Bu_{d,k}} + \boldsymbol{Dv_{w_k}}, \tag{5}$$

which is referred to as Model 1. We propose to consider another model, namely Model 2. In Model 2, we assume that the executed rotor speed $\boldsymbol{\Omega}_R$ can be measured with appropriate sensors. We estimate the cumulative thrust f_r using the quadratic rotor model and the measured $\boldsymbol{\Omega}_R$ and then compute the residual in the executed and the commanded thrusts as $f_{res} = f_c - f_r$. We augment Model 1 with f_{res} as an additional disturbance to obtain Model 2

$$\boldsymbol{\psi_{k+1}^e} = \boldsymbol{A\psi_k^e} + \boldsymbol{B}(\boldsymbol{u_{d,k}} + \boldsymbol{u_{res,k}}) + \boldsymbol{Dv_{w_k}}, \tag{6}$$

where $\boldsymbol{u_{res,k}} = [0,\ 0,\ 0, f_{res,k}]^T$. The two models (5) and (6) are learned using Extended Dynamic Mode Decomposition (EDMD) with l_1 regularization [15]. EDMD has been popular and utilized in a variety of platforms [22, 12, 23, 24]. Section III-B briefly reviews the Koopman operator theory and EDMD.

To reject the wind disturbances, we employ the learned dynamic model in MPC. We consider two MPC formulations. These two formulations differ in how wind velocities are predicted in the MPC optimization. In the first formulation, the stochastic component of the wind is ignored and thus the MPC formulations become fully deterministic. The first MPC formulation, namely K-MPC, we consider $\boldsymbol{v_{w_k}} = \bar{\boldsymbol{v}}_{\boldsymbol{w}}$. That is, we use the mean wind information to predict future wind velocities. The second MPC formulation follows a stochastic approach. This stochastic MPC, denoted by K-SMPC, considers $\boldsymbol{v_{w_k}} = \bar{\boldsymbol{v}}_{\boldsymbol{w}} + \tilde{\boldsymbol{v}}_{\boldsymbol{w_k}}$ with the mean component $\bar{\boldsymbol{v}}_{\boldsymbol{w}}$ and the stochastic component $\tilde{\boldsymbol{v}}_{\boldsymbol{w_k}}$. We assume that the covariance of $\tilde{\boldsymbol{v}}_{\boldsymbol{w_k}}$ is known. Then we formulate an SMPC with a tightening parameter to compensate for the stochastic $\tilde{\boldsymbol{v}}_{\boldsymbol{w_k}}$. Section III-C presents the details of the two MPC formulations.

Our method can be used in identifying wind-affected true dynamics of an off-the-shelf quadrotor and augmenting an existing controller to mitigate wind effects.

B. Model Identification with Koopman Operator

To approximate the Koopman operator, consider k discrete state measurements from datasets $\boldsymbol{X} = [x_1, x_2, \cdots, x_k]$, $\boldsymbol{Y} = [y_1, y_2, \cdots, y_k]$ and $\boldsymbol{U} = [u_1, u_2, \cdots, u_k]$ that can be arranged as snapshot pairs. We choose a basis of monomials with a total degree less or equal to p as the lifting function. The number of functions in the basis is equal to $N = (n+p)!/(n!\ p!)$ and the vector of the basis is $\boldsymbol{p(x)} = (p_1(x), ..., p_N(x))$. Then the lifted state is given by $\boldsymbol{\psi(x)} = (x, p_{n+1}(x)....., p_N(x))$. Lifting the states into N dimensional observables, we have snapshot pairs $\boldsymbol{\alpha_1}[k] = [\boldsymbol{\psi(x_k)}\ , \boldsymbol{u_k}]^T$, $\boldsymbol{\alpha_2}[k] = [\boldsymbol{\psi(y_k)},\ 0]^T$. Consider the least square approximation

$$\min_{\boldsymbol{M}} \|\boldsymbol{\alpha_2}[k] - \boldsymbol{M\alpha_1}[k]\|_2^2, \tag{7}$$

where the system matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are embedded in $\boldsymbol{M}$. For practical consideration where $k \gg N$ and to guard against over-fitting to noise, we utilize modified EDMD with l_1 regularization method [15]. The objective function becomes:

$$\min_{\vec{\boldsymbol{M}}} \|\vec{\boldsymbol{\Gamma_2}} - \vec{\boldsymbol{M}}\vec{\boldsymbol{\Gamma_1}}\|_2^2 + \lambda\|\vec{\boldsymbol{M}}\|_1. \tag{8}$$

here $\boldsymbol{\Gamma_1} = [\boldsymbol{\alpha_1}[1], \cdots \boldsymbol{\alpha_1}[k]]^T$, $\boldsymbol{\Gamma_2} = [\boldsymbol{\alpha_2}[1], \cdots, \boldsymbol{\alpha_2}[k]]^T$, λ is the weight of l_1 penalty term, and $\vec{\cdot}$ denotes a vectored version of each matrix with dimensions consistent with the

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problem. In the presence of wind disturbance $\boldsymbol{v_{w_k}}$, we consider

$$\boldsymbol{x_{k+1}} = \boldsymbol{f(x_k, u_{d,k}, v_{w_k})}. \tag{9}$$

With the disturbances in the form $\boldsymbol{V_w} = [\boldsymbol{v_{w_1}}, \boldsymbol{v_{w_2}}, \cdots, \boldsymbol{v_{w_k}}]$, we update the lifted snapshot pairs as $\boldsymbol{\alpha_1[k]} = [\boldsymbol{\psi(x_k)},\ \boldsymbol{u_{d,k}},\ \boldsymbol{v_{w_k}}]^T$. Then the updated optimization matrix $\boldsymbol{M}$ becomes

$$\boldsymbol{M} = \begin{bmatrix} \boldsymbol{A}_{N\times N} & \boldsymbol{B}_{N\times m} & \boldsymbol{D}_{N\times m_1} \\ \boldsymbol{\emptyset}_{(m+m_1)\times N} & \boldsymbol{\emptyset}_{(m+m_1)\times m} & \boldsymbol{\emptyset}_{(m+m_1)\times m1} \end{bmatrix} \tag{10}$$

resulting in the state space model of the form

$$\boldsymbol{\psi_{k+1}} = \boldsymbol{A\psi_k} + \boldsymbol{Bu_{d,k}} + \boldsymbol{Dv_{w_k}}. \tag{11}$$

Following the same optimization process, we obtain the models in (5) and (6), which are further used to design MPC controllers.

C. Koopman based MPC

Since the identified model is linear, a linear MPC can be designed to improve trajectory tracking. Considering the systems in (5) and (6), we first split the state into a deterministic error state $\boldsymbol{z_k}$ and a probabilistic error state $\boldsymbol{\zeta_k}$. Let $\boldsymbol{\nu_k}$ be the input for the deterministic part. We split the disturbance $\boldsymbol{v_{w_k}}$ into a deterministic part $\boldsymbol{\bar{v}_w}$ and a stochastic part $\boldsymbol{\tilde{v}_{w_k}}$. The $\boldsymbol{\bar{v}_w}$ is the mean wind, which is constant over time. Therefore, we have

$$\boldsymbol{\psi_k^e} = \boldsymbol{z_k} + \boldsymbol{\zeta_k} \tag{12a}$$

$$\boldsymbol{u_{d,k}} = \boldsymbol{\nu_k}, \tag{12b}$$

and the deterministic and stochastic dynamics are given by

$$\boldsymbol{z_{k+1}} = \boldsymbol{Az_k} + \boldsymbol{B_k\nu_k} + \boldsymbol{D\bar{v}_w} \tag{13a}$$

$$\boldsymbol{\zeta_{k+1}} = \boldsymbol{A\zeta_k} + \boldsymbol{D\tilde{v}_{w_k}}. \tag{13b}$$

We also consider a tracking error constraint given by

$$\boldsymbol{h_{lb_k}} \leq \boldsymbol{\psi_k^e} \leq \boldsymbol{h_{ub_k}} \tag{14a}$$

$$\boldsymbol{h_{lb_k}} \leq \boldsymbol{z_k} + \boldsymbol{\zeta_k} \leq \boldsymbol{h_{ub_k}}. \tag{14b}$$

The cost function used is given by

$$\boldsymbol{J} = (\boldsymbol{Cz_{n_p}})^T\boldsymbol{Q}(\boldsymbol{Cz_{n_p}}) + \sum_{K=0}^{(n_p-1)} (\boldsymbol{Cz_k})^T\boldsymbol{Q}(\boldsymbol{Cz_k}) + \boldsymbol{\nu_k}^T\boldsymbol{R\nu_k}. \tag{15}$$

The K-MPC focus only on the deterministic dynamics $\boldsymbol{z}_k$ and we assume $\boldsymbol{z_0} = \boldsymbol{\psi_0}$. The dynamics and constraints in K-MPC are given by

$$\boldsymbol{z_{k+1}} = \boldsymbol{Az_k} + \boldsymbol{B\nu_k} + \boldsymbol{D\bar{v}_w},\ k \in \{0, \cdots, n_p - 1\}, \tag{16a}$$

$$\boldsymbol{u_{lb}} \leq \boldsymbol{\nu_k} \leq \boldsymbol{u_{ub}},\ k \in \{0, \cdots, n_p - 1\}, \tag{16b}$$

$$\boldsymbol{h_{lb_k}} \leq \boldsymbol{z_k} \leq \boldsymbol{h_{ub_k}},\ k \in \{1, \cdots, n_p\}. \tag{16c}$$

We also formulate a stochastic MPC (SMPC) to incorporate the probabilistic information. First, we introduce a probabilistic constraint (chance constraint) that will bound the constraint violation probability depending on a risk parameter, β, and update the state constraint as a chance constraint

$$Pr[\boldsymbol{h_{lb_k}} \leq \boldsymbol{z_k} + \boldsymbol{\zeta_k} \leq \boldsymbol{h_{ub_k}}] \geq \beta. \tag{17}$$

The higher the risk parameter is chosen, the lower the risk allowed. The chance constraint (17) is not a deterministic expression. Here, we split the deterministic and probabilistic parts and include a tightening parameter γ to separate the deterministic and probabilistic parts

$$\boldsymbol{h_{lb_k}} + \boldsymbol{\gamma_k} \leq \boldsymbol{z_k} \leq \boldsymbol{h_{ub_k}} - \boldsymbol{\gamma_k} \tag{18a}$$

$$Pr[-\boldsymbol{\gamma_k} \leq \boldsymbol{\zeta_k} \leq \boldsymbol{\gamma_k}] = \beta. \tag{18b}$$

We assume that the mean of the disturbance, $\boldsymbol{\bar{v}_w}$, is known and $\boldsymbol{\tilde{v}_{w_k}}$ follows a zero-mean Gaussian distribution with covariance $\boldsymbol{\Sigma_{v_w}}$, i.e., $\boldsymbol{\tilde{v}_{w_k}} \sim N(0, \boldsymbol{\Sigma_{v_w}})$. The $\boldsymbol{\Sigma_{v_w}}$ and $\boldsymbol{\bar{v}_w}$ may be available from a weather tower. The probability distribution of the error state follows $\boldsymbol{\zeta_k} \sim N(0, \boldsymbol{\Sigma_k^\zeta})$ with $\boldsymbol{\Sigma_0^\zeta} = 0$. We compute the prediction error covariance as $\boldsymbol{\Sigma_{k+1}^\zeta} = \boldsymbol{A\Sigma_k^\zeta A^T} + \boldsymbol{D\Sigma_{v_w}D^T}$. Based on the cumulative distribution function of a Gaussian distribution we obtain $\beta = erf\left[\frac{\gamma_k}{\sigma\sqrt{2}}\right]$.

Hence, the deterministic expression for the tightening parameter is $\boldsymbol{\gamma_k} = \sqrt{2\boldsymbol{v_k^\zeta}}\ erf^{-1}(\beta)$, where $\boldsymbol{v_k^\zeta} = diag(\boldsymbol{\Sigma_k^\zeta})$ contains the diagonal elements of the error covariance matrix. The constraints in the K-SMPC then become

$$\boldsymbol{h_{lb_k}} + \boldsymbol{\gamma_k} \leq \boldsymbol{z_k} \leq \boldsymbol{h_{ub_k}} - \boldsymbol{\gamma_k},\ k \in \{1, \cdots, n_p\}, \tag{19a}$$

$$\boldsymbol{\gamma_k} = \sqrt{\boldsymbol{2v_k^\zeta}}\ \boldsymbol{erf^{-1}(\beta)},\ k \in \{1, \cdots, n_p\}, \tag{19b}$$

$$\boldsymbol{\Sigma_{k+1}^\zeta} = \boldsymbol{A\Sigma_k^\zeta A^T} + \boldsymbol{D\Sigma_{v_w}D^T},\ k \in \{1, \cdots, n_p\}. \tag{19c}$$

Similarly to [25], the MPC optimization can be solved from a dense form.

IV. SIMULATION RESULTS

A. Data Collection

To collect the learning dataset, we use two reference trajectories of 14 seconds that are tracked via the LQR. Each trajectory is repeated four times with different perturbations on the high-level control $\boldsymbol{u_h}$ and wind velocity. We use a realistic wind field that considers complex highly coherent turbulent structures from Large-Eddy Simulations [26]. The mean wind velocity is $\boldsymbol{\bar{v}_w} = [2.7032\ , 1.7566\ , 0.0124]^T$ m/s. The state, actuation, and wind datasets are captured at 10 Hz. Two validation data sets are also generated in the same way with different wind and random perturbations on $\boldsymbol{u_h}$.

B. System Identification

The error dynamics of the system are then learned with EDMD. The high-dimensional Koopman model reduces the prediction error by 76% in the validation set. To generate datasets for models 1 and 2 in (5) and (6), respectively, the same pre-generated random perturbations and the same reference trajectory are used apart from the residual force. This enables us to observe the effects of the residual force in the prediction and control approach. The approximation utilizes a set of 1126 pairs of snapshots with l_1 regularization parameter of 20. The total dimension of the lifted state space is $N = 55$, consisting of monomials of degree $p = 2$. The prediction in a validation dataset is illustrated in Fig. 2. Model prediction accuracy is evaluated against a validation

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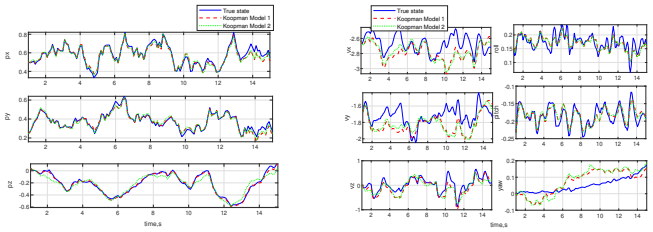


Fig. 2. Position, velocity and orientation prediction with the two Koopman models.

set that has not been used in the training. Plots in Fig. 2 show the actual and predicted trajectory from models 1 and 2. The average rms error of position from the prediction of both models is provided in Table I. Both models perform well in predicting the position.

TABLE I

RMSE ERROR FOR PREDICTION

	px	py	pz
Model 1	0.055441	0.050398	0.023126
Model 2	0.050389	0.043464	0.007532

C. Trajectory Tracking Analysis

To investigate the closed-loop control performance of the models, we implement two controllers designed in Section III-C. The tracking performance of the controller is assessed with Monte Carlo simulations with different sets of LES wind. The penalty matrices $\boldsymbol{Q}$ and $\boldsymbol{R}$ used in the LQR and in the MPC are the same for both controllers. We set the prediction horizon, $n_p = 5$. The value of the tightening parameter $\boldsymbol{\gamma}$ depends on the variance $\boldsymbol{\Sigma_{v_w}} = diag([0.5434, 0.4197, 0.0064])$ and the risk parameter $\beta = 0.99$. Increasing β increases $\boldsymbol{\gamma}$, as $\boldsymbol{\gamma}$ is proportional to the value of inverse error function of β (see (19b)).

TABLE II

COMPARISON OF EUCLIDEAN NORM OF THE TRACKING ERROR

Controller	Model 1		Model 2	
	value	% reduction	value	% reduction
LQR	0.6812	-	-	-
LQR+K-MPC	0.6288	7.69%	0.6136	9.92%
LQR+K-SMPC	0.5611	17.63%	0.4656	31.65%

TABLE III

ROOT MEAN SQUARE ERROR (RMSE) COMPARISON

Controller	rmse x		rmse z	
	Model 1	Model 2	Model 1	Model2
LQR	0.6029	-	0.1690	-
LQR+K-MPC	0.5758	0.5547	0.0683	0.0550
LQR+K-SMPC	0.4795	0.2604	0.0468	0.0533

We compare the tracking errors against the baseline LQR controller. Integrating Koopman MPC improves the tracking

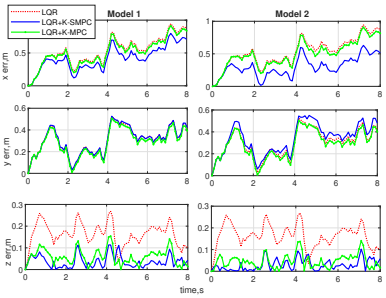


Fig. 3. Trajectory tracking error for model 1 and model 2. Model 2 enables reducing the error in x and z directions significantly. In both cases, augmenting LQR with MPC based on the learned models improves the accuracy.

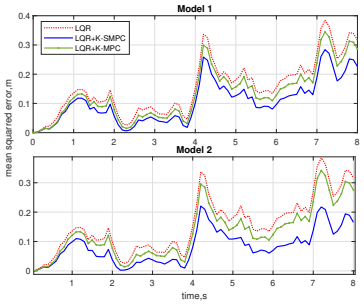


Fig. 4. Mean squared error (mse) over five Monte Carlo simulations throughout the trajectory using model 1 and model 2.

performance in every case for both models. Fig. 3 shows the absolute tracking error with model 1 and model 2. While the tracking error is reduced in both models, model 2 performs better in reducing the error. This is also prominent from Fig. 4 that shows the mean squared error with different controller combinations of both models. It should be noted that the error at each timestamp is dominated by the time-variant turbulent wind, which explains the increase and decrease in absolute and mean squared error at about $t = 4.3$ seconds and $t = 5.8$ seconds consecutively. However, the constraints were satisfied throughout the simulations and do not incur any divergence. The Euclidean norm of the tracking error with model 1 and model 2 in Table II shows significant improvement when we augment the existing LQR with MPC. Both K-MPC and K-SMPC exhibit better performance in terms of reducing error. However, model 2 exhibits consistent improvement. With K-SMPC it reduces error by 31.65%, which is the best performance among all combinations. Table III shows the rms tracking error along x, and z directions. With model 2, the rms error is significantly reduced in x and z directions by as low as 56% and 68%. Augmenting LQR with SMPC also enables the system to respond faster to the disturbance.

V. CONCLUSIONS AND FUTURE WORK

In this work, Koopman based data-driven modeling and control were successfully applied to model error and residual dynamics of a quadcopter flying in a turbulent wind. The results show significant improvement in reducing tracking

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error when the existing controller is augmented with a MPC based on the learned Koopman models. Future work will explore the proposed methodology with sensor integration and bi-linear modeling.

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